

Experiment 2

Aim : Write a program to implement Group Communication.

Theory :

- In distributed system, it is necessary to have support for sending data to multiple receivers.
- This type of communication is called multicast communication.
- To support for ~~see~~ multicast communication, many network-level and transport-level solutions have been implemented.
- The issues in these solutions were to setup the path to disseminate the information.
- Alternative to these solutions is application level multicast techniques as peer to peer solutions are usually deployed at application layer.
- It now became easier to setup communication paths.
- Multicast messages offer good solution to build distributed system with following characteristics:

- 1) Fault tolerance based on replicated service
- Replicated service is available on group of servers. Client requests are multicast to all these servers in group. Each server performs the same operation on this request.

2) Finding discovery servers in spontaneous networking

⇒ Servers and clients can use multicast messages to find discovery services to register their interfaces of other services in the distributed system.

3) Better performance through replicated data

⇒ Replication improves performance. If in some cases, replica of data is placed in client machine then changes in data item is multicast to all the processes managing the replicas.

4) Event notification

⇒ Multicast to group to notify if some event occurs.

→ Group communication is essential in distributed systems for efficient coordination, fault tolerance, and consistency.

→ Mechanisms like multicast, publish-subscribe and gossip protocols enable's scalable and reliable communication between multiple nodes.

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